SIMATS ENGINEERING

ASSESSMENT TOOL 8 - MEDIUM

(SCENARIO-BASED QUESTIONS)

COURSE CODE: ITA04

COURSE NAME: STATISTICS WITH R PROGRAMMING

COURSE OUTCOME COVERED

CO: Construct and manipulate data frames using reshaping, merging, and file I/O operations in R to prepare structured datasets for analysis.

(Bloom Level -BL3- BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 20 **Total Marks: 100**

ANNEXURE A

SCENARIO BASED QUESTIONS QUESTIONS

Code of Conduct:

*I, **Savitha R / 192421368** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Savitha R

SCENARIO BASED QUESTIONS

Frames (Data Frames)

Question 1
Scenario: Smart Manufacturing
Task: Analyze sensor dataframe from 5 machines to identify defective batches using filtering, grouping and merging maintenance logs.
Expected Answer: Create data frames, merge(), subset(), aggregate(); identify machine with highest defect rate.

```
> # Sensor data
> sensor <- data.frame(
+     MachineID=c(1,2,3,4,5),
+     Defects=c(2,5,1,4,6)
+ )
>
> # Maintenance data
> maint <- data.frame(
+     MachineID=c(1,2,3,4,5),
+     Status=c("OK","Repair","OK","Repair","Repair")
+ )
>
> # Merge
> data <- merge(sensor, maint, by="MachineID")
>
> # Filter defective machines
> subset(data, Defects > 3)
  MachineID Defects Status
2         2       5 Repair
4         4       4 Repair
5         5       6 Repair
>
> # Defect rate
> aggregate(Defects ~ MachineID, data, mean)
  MachineID Defects
1         1       2
2         2       5
3         3       1
4         4       4
5         5       6
```

Question 2

Scenario: Hospital

Task: Merge patient admissions and lab data to find delayed diagnosis.

Expected Answer: Use merge(), handle NA, summarize delays.

```
> patient <- data.frame(
+     ID=c(1,2,3),
+     Delay=c(2,NA,5)
+ )
>
> lab <- data.frame(
+     ID=c(1,2,3),
+     Result=c("P","N","P")
+ )
>
> data <- merge(patient, lab, by="ID")
>
> data$Delay[is.na(data$Delay)] <- 0
>
> summary(data$Delay)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   1.000   2.000   2.333   3.500   5.000
```

Question 3

Scenario: Traffic

Task: Reshape traffic dataframe to compare peak-hour congestion across junctions.

Expected Answer: Use reshape, aggregate and visualize.

```
> traffic <- data.frame(
+     Junction=c("A","B","C"),
+     Morning=c(120,100,140),
+     Evening=c(180,150,200)
+ )
>
> long <- reshape(
+     traffic,
+     varying=c("Morning","Evening"),
+     v.names="Vehicles",
+     timevar="Time",
+     times=c("Morning","Evening"),
+     direction="long"
+ )
>
> aggregate(Vehicles ~ Junction, long, mean)
  Junction Vehicles
1        A      150
2        B      125
3        C      170
```

Question 4

Scenario: Retail

Task: Detect inconsistent sales records after combining branch data frames.
Expected Answer: Merge, remove duplicates, validate totals.

```
> branch1 <- data.frame(
+     ID=c(1,2,3),
+     Sales=c(100,200,300)
+ )
>
> branch2 <- data.frame(
+     ID=c(3,4,5),
+     Sales=c(300,400,500)
+ )
>
> sales <- rbind(branch1, branch2)
>
> sales <- sales[!duplicated(sales), ]
>
> sum(sales$Sales)
[1] 1500
```

Question 5
Scenario: University
Task: Evaluate student performance by merging attendance and marks.
Expected Answer: Merge, compute weighted score, rank students.

```
> attendance <- data.frame(
+     ID=c(1,2,3),
+     Attendance=c(90,80,95)
+ )
>
> marks <- data.frame(
+     ID=c(1,2,3),
+     Marks=c(85,75,95)
+ )
>
> student <- merge(attendance, marks, by="ID")
>
> student$Score <- 0.4*student$Attendance +
+     0.6*student$Marks
>
> student$Rank <- rank(-student$Score)
>
> print(student)
  ID Attendance Marks Score Rank
1  1         90    85    87    2
2  2         80    75    77    3
3  3         95    95    95    1
```

Files in R Programming

Question 1

Scenario: IoT CSV

Task: Read daily CSV files, combine them, remove missing values and detect anomalies.

Expected Answer: Use read.csv(), list.files(), rbind(), na.omit().

setwd("C:/Users/DELL/OneDrive/Desktop/Federated Learning")

list.files(pattern = "*.csv")

data <- read.csv("waterQuality1.csv")

head(data)

data <- na.omit(data)

subset(data, ammonia > 20)

```
4   0.004         9.12  1.72      0.02  0.45    0.05         1
5   0.003        16.90  2.41      0.02  0.06    0.02         1
6   0.006        27.17  5.42      0.08  0.19    0.02         1
> data <- na.omit(data)
> subset(data, ammonia > 20)
   aluminium ammonia arsenic barium cadmium chloramine chromium copper flouride bacteria viruses  lead nitrates
1       1.65    9.08   0.040   2.85   0.007       0.35     0.83   0.17     0.05     0.20   0.000 0.054    16.08
2       2.32   21.16   0.010   3.31   0.002       5.28     0.68   0.66     0.90     0.65   0.650 0.100     2.01
5       0.92   24.33   0.030   0.20   0.006       2.67     0.69   0.57     0.61     0.13   0.001 0.117     6.74
7       2.36     5.6   0.010   1.35   0.004       1.28     0.62   1.88     0.33     0.13   0.007 0.021    18.60
9       0.60   24.58   0.010   0.71   0.005       3.14     0.77   1.45     0.98     0.35   0.002 0.167    14.66
11      3.27     3.6   0.001   2.69   0.005       5.75     0.15   0.60     1.29     0.04   0.008 0.145     8.47
12      1.35   21.96   0.040   0.84   0.002       0.10     0.76   0.17     0.58     0.52   0.520 0.011    18.40
14      4.93   23.98   0.040   3.05   0.008       0.70     0.51   1.35     1.07     0.70   0.700 0.101     1.16
19      4.88   26.94   0.020   0.36   0.001       1.21     0.68   0.71     0.99     0.75   0.750 0.071     0.31
22      1.15    8.12   0.020   0.97   0.007       3.47     0.65   1.51     1.46     0.58   0.580 0.061     8.96
24      4.32   20.64   0.030   2.60   0.008       7.24     0.61   1.23     1.44     0.56   0.560 0.012     9.42
25      2.36   27.05   0.010   0.68   0.003       4.07     0.13   1.34     0.29     0.96   0.960 0.167    15.05
26      3.31   22.07   0.030   0.46   0.001       7.22     0.73   1.05     1.00     0.25   0.007 0.109     1.92
27      1.82    6.81   0.010   0.85   0.006       2.55     0.25   1.09     1.35     0.16   0.002 0.031    16.99
```

Question 2

Scenario: Manufacturing Logs

Task: Write cleaned production report to CSV and verify integrity.

Expected Answer: Use write.csv(), compare row counts.

```
> # Set working directory
> setwd("C:/Users/DELL/OneDrive/Desktop/Federated Learning")
>
> # Read CSV
> data <- read.csv("waterQuality1.csv")
>
> # Write cleaned report
> write.csv(data, "production_report.csv", row.names = FALSE)
>
> # Verify row count
> print(nrow(data))
[1] 7999
```

Question 3

Scenario: Network Logs

Task: Import multiple log files and identify failed login patterns.

Expected Answer: Read files, combine, filter failures.

```
> setwd("C:/Users/DELL/OneDrive/Desktop/Federated Learning")
>
> logs <- read.csv("waterQuality1.csv")
>
> subset(logs, is_safe == 0)
   aluminium ammonia arsenic barium cadmium chloramine chromium copper flouride bacteria viruses  lead nitrates
3       1.01   14.02   0.040   0.58   0.008       4.24     0.53   0.02     0.99     0.05   0.003 0.078    14.16
7       2.36     5.6   0.010   1.35   0.004       1.28     0.62   1.88     0.33     0.13   0.007 0.021    18.60
8       3.93   19.87   0.040   0.66   0.001       6.22     0.10   1.86     0.86     0.16   0.005 0.197    13.65
11      3.27     3.6   0.001   2.69   0.005       5.75     0.15   0.60     1.29     0.04   0.008 0.145     8.47
19      4.88   26.94   0.020   0.36   0.001       1.21     0.68   0.71     0.99     0.75   0.750 0.071     0.31
24      4.32   20.64   0.030   2.60   0.008       7.24     0.61   1.23     1.44     0.56   0.560 0.012     9.42
25      2.36   27.05   0.010   0.68   0.003       4.07     0.13   1.34     0.29     0.96   0.960 0.167    15.05
28      3.42     2.4   0.001   2.80   0.003       2.87     0.73   0.27     0.53     0.44   0.002 0.110    13.73
30      4.57   25.84   0.010   3.04   0.002       2.78     0.72   0.46     1.41     0.08   0.006 0.049     9.64
34      1.63   15.75   0.030   2.54   0.008       4.25     0.74   1.35     1.24     0.12   0.006 0.193    17.29
35      0.01   29.29   0.001   2.93   0.007       7.75     0.68   1.09     0.00     0.80   0.800 0.091    17.33
39      2.51   13.08   0.040   1.58   0.002       2.86     0.87   1.88     0.40     0.13   0.004 0.085    11.96
```

Question 4

Scenario: Weather Data

Task: Read text and CSV formats and compare averages.

Expected Answer: Use read.table(), read.csv(), summary().

```
> # Read CSV file
> csvData <- read.csv("weather.csv")
>
> # Display summaries
> summary(textData)
  Temperature        Rainfall    
 Min.   :28.00   Min.   :15.00  
 1st Qu.:29.50   1st Qu.:17.25  
 Median :30.50   Median :19.00  
 Mean   :30.25   Mean   :19.50  
 3rd Qu.:31.25   3rd Qu.:21.25  
 Max.   :32.00   Max.   :25.00  
> summary(csvData)
  Temperature        Rainfall    
 Min.   :28.00   Min.   :15.00  
 1st Qu.:29.50   1st Qu.:17.25  
 Median :30.50   Median :19.00  
 Mean   :30.25   Mean   :19.50  
 3rd Qu.:31.25   3rd Qu.:21.25  
 Max.   :32.00   Max.   :25.00  
```

Question 5

Scenario: Smart City

Task: Export analyzed pollution report for management.

Expected Answer: Analyze then write.csv() final report.

```
> setwd("C:/Users/DELL/OneDrive/Desktop/Federated Learning"
>
> data <- read.csv("pollution.csv")
>
> summary(data)
     Area                AQI
 Length:4           Min.   : 80.00
 Class :character   1st Qu.: 91.25
 Mode  :character   Median :107.50
                    Mean   :111.25
                    3rd Qu.:127.50
                    Max.   :150.00
>
> write.csv(data,
+           "pollution_report.csv",
+           row.names = FALSE)
>
> print("Report exported successfully")
[1] "Report exported successfully"
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively	Applies relevant concepts correctly	Applies basic concepts but with	Fails to apply appropriate concepts

		to solve complex real-world problems	with minor errors	limited accuracy	
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification